

Yang Xue

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EDUCATION

Queen's University

Bachelor's of Health Sciences

Kingston, ON

Expected Graduation May 2030

John Fraser Secondary School

AP Student

EXPERIENCE

Technology Executive

September 2023 - June 2026

Student Activity Council

Mississauga, ON

- Eliminated ticket fraud risk for the school's biggest event, replacing all physical tickets by engineering a **digital wallet system** that processed **\$15,000+** in sales.
- Centralized school communication and enhanced event experiences for **1,000+ active users** by programming multiple websites such as an application dashboard, elections page, and voting dashboard.

Teaching Executive

September 2024 - June 2026

JFSS HOSA

Mississauga, ON

- Coached HOSA members to **multiple top-10 national placements** by designing structured study programs and mock competition sessions
- Grew club application volume by **83%** by overhauling the recruitment process and expanding outreach across the school

Vice President of Logistics

June 2024 – April 2026

HackCanada

Waterloo, ON

- Coordinated logistics for a **48-hour** hackathon with **1000+ participants**, ensuring smooth operations across sponsors, scheduling, and event execution
- Managed distribution and organization of **\$20,000+** in prizes and sponsor resources for attendees and competition winners

ACHIEVEMENTS

National Champion & International Finalist | HOSA Veterinary Science

March 2025 – June 2025

- Placed **1st nationally** and **8th** in a veterinary science competition against **200+ national** and **212 international** competitors

PROJECTS

Auditory-Affective Compensation in Capgras Syndrome | 1st Place

August 2026

- Led a **month-long neuroscience research study** with **4 team members** investigating whether voice familiarity can restore visual-affective deficits in post-stroke Capgras syndrome, placing **1st at StemBuddies Research Symposium**
- Designed a counterbalanced experimental protocol with three conditions (visual-only, visual + voice, simulated conversation) using **skin conductance sensors**, **PSYRATS**, and **Wisconsin Card Sorting Test** to measure autonomic arousal and delusional conviction
- Conducted a **PRISMA-guided systematic review**, synthesizing 17 studies to support hypothesis that auditory-affective engagement can modulate belief-evaluation networks

Methanogenic Interception of TMA by *M. luminyensis* to Reduce TMAO-Driven Cardiovascular Risk

July 2026

- Proposed **ArchaeBiotic**, a novel oral capsule treatment, in a **6-hour** microbiology research competition, designed to divert trimethylamine (TMA) toward methanogenesis instead of its conversion into cardiovascular-disease-linked TMAO
- Designed a dual-action delivery mechanism combining a pH-sensitive hydrogel shell for colon-targeted release of *M. luminyensis* with a prebiotic seaweed fibre to sustain hydrogen-dependent methanogenesis
- Synthesized findings from **9 primary sources** spanning gut microbiology, drug delivery, and cardiovascular disease to support the treatment's mechanism, safety rationale, and cost-effectiveness
- Collaborated with 2 peers under strict time constraints to research, write, and finalize the proposal